Alexander Anderson

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EDUCATION

University of Portland, Portland, OR **May 2026**
Bachelor of Science, Electrical Engineering (GPA 3.15) | Minor, Artificial Intelligence (GPA 3.5) | Merit Scholarship

SKILLS

Data & Software: Python, SQL, Snowflake (Cortex, AI_EXTRACT, Streamlit), dbt, Flask REST APIs, OpenCV, Docker, Git, Bitbucket/GitLab, Linux
Embedded & Hardware: C, C++, C#, MATLAB, Verilog/VHDL, KiCad, SolidWorks, Fusion 360, Jetson, PXIe test automation, PCB design, UART/I2C/CAN, oscilloscope & bench debugging, 3D printing
AI & ML: neural networks, reinforcement learning, A*/minimax search, leave-one-out cross-validation, computer vision, LLM-based data extraction
Interests: Sensor technology for chronic pain and neuromodulation, AI in medical devices, fMRI

PROFESSIONAL EXPERIENCE

Engineering Intern, Neuro Clinical Technology Group **June 2026 – Sep 2026**
BIOTRONIK (Microsystems Engineering Inc.) | *Lake Oswego, OR*

- Designed and built an end-to-end sleep-data characterization platform for an implantable neurostimulator's sleep-detection algorithm: Snowflake + dbt pipeline (152/152 tests passing), Flask REST API (~43 routes, 1,268 unit tests), and a Streamlit-in-Snowflake review app now used by the algorithm owner.
- Built a device-to-participant crosswalk that joins four incoming sources (implant actigraphy, wrist watch, Apple Health, paper diary) onto a common night/epoch grid; added participant-ID masking and a reviewer-corrections flow into the marts.
- Automated handwritten sleep-diary extraction by combining Snowflake AI_EXTRACT with a custom OpenCV AM/PM mark classifier: 91% overall and 98% on clean marks vs. 70 hand-verified cells.
- Built a fail-closed ML layer (per-night feature extraction, rules-based labeling, multi-model evaluation with leave-one-tester-out CV) staged for clinical reference data, and Ask BioSleep, a Snowflake Cortex assistant grounded in ~101 reviewed, cited answers.
- Root-caused pipeline defects: corrected a degenerate-night rate from 70% to ~25% (duplicate ingestion, non-wear gaps) and traced device clock-offset errors up to ±7 h (cross-source agreement 0.657 to 0.714 when corrected).
- Reverse-engineered a proprietary actigraphy data format, recovering ~82 device-days of missed history. Replayed 59 algorithm debug nights exactly and characterized 60 config parameters over 89 clean nights (5 sensitive, 13 inert, 16 dead).
- Set up Bitbucket and a Git-linked dbt project, wrote 10 runbooks and a handover guide, and presented the platform to the clinical engineering team.

Electrical & Controls Engineer, NASA Lunabotics Capstone **Sep 2025 – May 2026**
University of Portland

- Designed a sealed, dust-proof electrical enclosure (gaskets, cable glands, 3D-printed mounts) and a 24 V power-distribution and current-sensing network with soft-start sequencing to prevent brownouts.
- Built autonomous navigation using A* with dynamic replanning and AprilTag-based localization; implemented gradual motor start/stop control and multi-camera CSV logging on the Jetson.
- Implemented UART/CAN communication between microcontroller and Jetson for motor control and sensor feedback; produced schematics, wiring diagrams, and CAD for NASA SRR, PDR, and CDR reviews.

Electrical Engineering Intern (Quality & Reliability) **May 2024 – Aug 2024**
Microsystems Engineering Inc. | *Portland, OR*

- Developed and deployed embedded PXIe test automation (C, C#, LabWindows/CVI) for continuity and isolation checks on de-processed ICs, integrating SMUs, digital I/O, relay matrices, and DAQ modules.
- Designed test sequences with relay routing, power-sequencing safety, and spec-based fault flagging in a custom GUI; authored startup and troubleshooting documentation.
- Supported IC reliability work: packaging analysis, wire-bond integrity testing, and pad-type classification for implantable medical-device SoCs.

Undergraduate Student Researcher, Dr. Kimball Biomedical Lab **Jun 2023 – Sep 2023**
University of Portland

- Designed and fabricated a full-scale lower-body negative pressure (LBNP) chamber, including CAD, a sliding patient rail, and a custom Fusion 360 PCB for pressure sensing and safety interlock control.
- Integrated pressure transducers, microcontroller DAQ, and emergency cutoff relay; validated performance from +50 to −100 mmHg. Co-authored a NASA-referenced research proposal and safety protocols.

Student Engineer, Engineering World Health **Jul 2022 – Aug 2022**
Dominican Republic

- Repaired, calibrated, and documented medical devices (infusion pumps, electrosurgical units, oxygen concentrators, vital-signs monitors) in low-resource hospitals, with board-level troubleshooting.
- Created bilingual documentation and repair logs to support technician training.

ACADEMIC PROJECTS & RESEARCH

fMRI and MRI Signal Processing Research **Aug 2025 – Dec 2025**

- Analyzed MRI/fMRI datasets with SPM12, MATLAB, and EEGLAB (motion correction, co-registration, statistical contrast mapping); wrote Python/MATLAB scripts for BOLD response curves and activation maps.

Artificial Intelligence and Autonomous Robotics Coursework **Jan 2025 – Dec 2025**

- Implemented A*, minimax, neural-network, reinforcement-learning, and Soar agents; programmed autonomous robots in Python and OpenCV with state-machine control, localization, and state estimation.

Verilog Digital Systems and Embedded Systems Design **Aug 2023 – Dec 2023**

- Designed multi-module Verilog systems with simulation and waveform validation, and a microcontroller system using timers, interrupts, UART/I2C, and ADCs.

LEADERSHIP

IEEE, University of Portland Chapter: Treasurer/Secretary (Jan 2022 – 2024), Member (Jan 2022 – present) | **Robotics Club:** Member (Sep 2025 – Nov 2025)